# Final Results: Computational Modeling of (TAPVR)

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# Motivation and Significance

## Core clinical problem

In TAPVR, oxygen-rich pulmonary venous blood returns to the right side instead of the left atrium.

Systemic blood flow depends on atrial-level communication, and obstruction can rapidly worsen pressure buildup and oxygen delivery.

### Why it matters medically

Critical neonatal congenital defect
Can cause cyanosis and poor systemic oxygen delivery
Obstructed cases are much more unstable

### Why modeling helps

Links abnormal anatomy to pressure, flow, and oxygen delivery
Tests “what-if” changes that are hard to isolate clinically
Compares normal, PAPVR, TAPVR, and obstructed TAPVR

### What our model enables

Quantify oxygen delivery index relative to normal
Sweep obstruction and ASD restriction parameters
Identify which features drive hemodynamic severity

Goal: explain why TAPVR becomes dangerous by simulating how abnormal venous routing, obstruction, and atrial shunting affect systemic oxygen delivery.

# Project Progression / Timeline

## Presentation 1

*Project idea / proposed model*

Defined TAPVR problem

Proposed lumped-parameter model

Introduced key variables:

f_anom, Rpv,anom, RASD

Identified outputs:

pressure, flow, oxygenation

## Presentation 2

*Preliminary results*

Built first TAPVR/PAPVR code

Compared normal, PAPVR, TAPVR

Added obstruction and ASD sweeps

Found expected physiological trends

Identified limitations in pressure scale

## Final Presentation

*Final results / refined model*

Used neonate/infant scaling

Tuned parameter magnitudes

Added O2 delivery index

Normalized results to % of normal

Refined final conclusions and limitations

# Innovation of Modeling Approach

- Tunable TAPVR/PAPVR routing: uses $f_{a\square o\square}$ to switch from normal circulation → PAPVR → TAPVR
- Separate disease mechanisms: tests pulmonary venous obstruction ($R^{P\square}$,anom) and ASD restriction ($R^{ASD}$) independently
- Neonate-scaled final model: uses infant-scaled heart rate, body mass, flows, and volumes instead of a generic adult model
- Oxygen delivery index: combines systemic saturation and aortic output, giving a better severity metric than saturation alone
- Fast parameter-sweep framework: compares multiple “what-if” scenarios without needing patient-specific CFD geometry
- Key innovation: a simple circuit model that links TAPVR anatomy to measurable hemodynamic severity.

# What changed since preliminary results?

### 1. Infant-scaled model
Changed default profile to a neonate/infant parameter set: HR = 140 bpm, body mass = 3.5 kg, smaller flow/volume scale.

### 2. Tuned pressure scale
Retuned cardiovascular resistances and chamber parameters to avoid the unrealistic LV pressure magnitudes seen in the preliminary model.

### 3. Added $O_2$ delivery index
Combined saturation and aortic output into one metric: $O_2$ Index = QAo × Ssys, then normalized to the normal case.

### 4. Sensitivity sweeps
Kept obstruction and ASD restriction sweeps to show what the model enables beyond a single simulation.

### 5. Literature comparison
Compared our lumped-parameter model against existing TAPVC/TAPVR modeling approaches, including CFD, to clarify how our model is different.

### 6. Clear final outputs
Focused final slides on $O_2$ delivery, Ppv, Psa, and QAo instead of over-emphasizing pressure-volume loops.

# Model Setup: LV model to Congenital Circulation

- Expanded the model to include RA, LA, pulmonary venous reservoir, systemic arteries, and atrial shunting.
- Added a switch for normal circulation, PAPVR, TAPVR, and obstructed TAPVR.
- Pulmonary veins return oxygenated blood to the right side instead of the left atrium
- Survival depends on an atrial-level opening, such as an ASD
- Goal: model how TAPVR affects pressure, flow, and oxygen delivery

- PAPVR/TAPVR switch: $f_{a\square o\square}$ = 0 normal, 0.5 PAPVR-like, 1.0 TAPVR-like

- Obstruction is modeled by increasing $R^{P\square}$,anom, the resistance of the anomalous pulmonary venous pathway.

# Model used in the Code

Core Equations

Pressure–volume: $P_i = (V_i - V_{0,i}) / C_i$

Flow: $Q_{i \to \square} = (P_i - P_\square) / R_{i\square}$

Volume balance: $dV_i/dt = Q_{i\square} - Q_{ou\square}$

$O_2$ saturation: $S_i = M_i / V_i$

$O_2$ balance: $dM_i/dt = Q_{i\square}S_{i\square} - Q_{ou\square}S_i$

Interpretation: the model conserves blood volume and oxygen content while rerouting pulmonary venous return from LA to RA.

TAPVR/PAPVR routing equations

$Q^{P\square \to RA} = f_{a\square o\square} \cdot (P^{P\square} - P^{RA}) / R^{P\square,anom}$

$Q^{P\square \to LA} = (1 - f_{a\square o\square}) \cdot (P^{P\square} - P^{LA}) / R^{P\square,norm}$

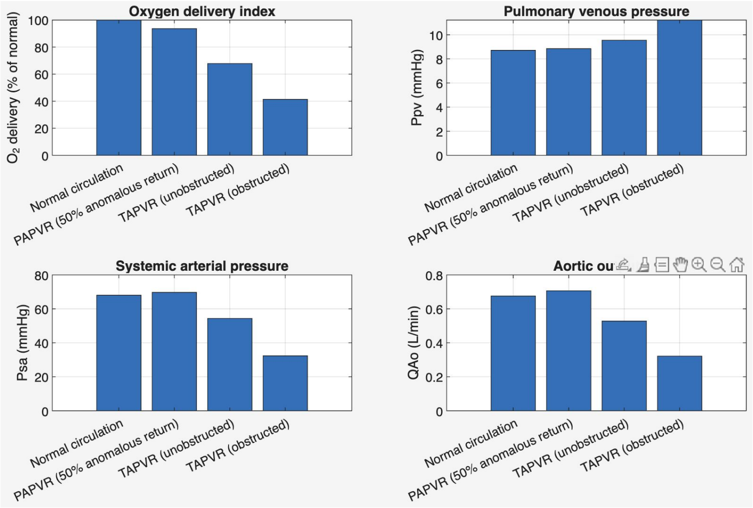
$Q^{AsD} = (P^{RA} - P^{LA}) / R^{AsD}$

Parameter meaning:

- $f_{a\square o\square}$ = fraction of anomalous pulmonary venous return
- $R^{P\square,anom}$ = obstruction severity
- $R^{AsD}$ = atrial opening restriction

# Main Comparison

| Case | $O_2$ %Norm | Ppv | Psa | QAo |
|---|---|---|---|---|
| Normal | 100.0 | 8.71 | 68.05 | 0.676 |
| PAPVR 50% | 93.7 | 8.85 | 69.72 | 0.706 |
| TAPVR | 67.9 | 9.55 | 54.37 | 0.528 |
| Obstructed | 41.4 | 11.22 | 32.36 | 0.322 |




- Trend: Normal > PAPVR > TAPVR for systemic saturation and aortic output.
- PAPVR causes mild/moderate reduction in $O_2$ delivery despite similar aortic output.
- TAPVR reduces systemic pressure, output, and $O_2$ delivery.
- Obstructed TAPVR is most severe: $O_2$ delivery falls to 41.4% of normal.

# Parameter Sweep

$R_{pv,anom}$ sweep: 8 → 30 mmHg/(L/min).

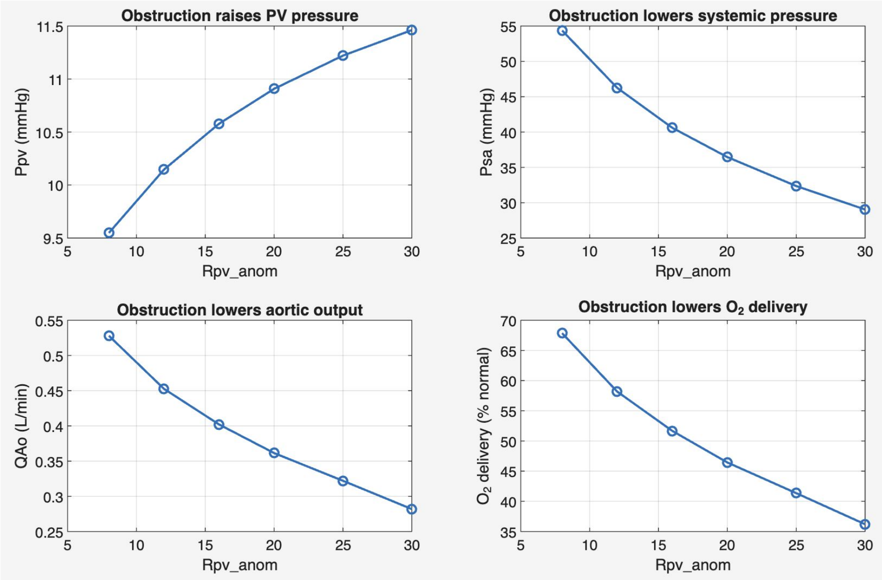
Pulmonary venous pressure rises: 9.55 → 11.46 mmHg.

Systemic arterial pressure falls: 54.37 → 29.04 mmHg.

Aortic output falls: 0.528 → 0.282 L/min.

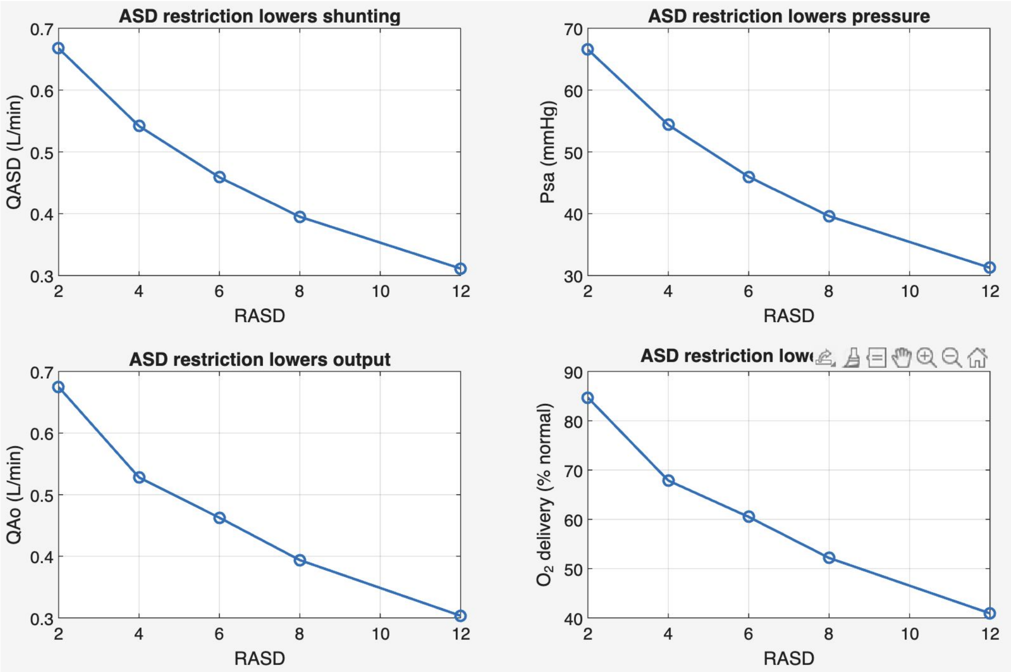
$O_2$ delivery falls: 67.9% → 36.2% of normal.

- Takeaway: Obstruction is a major driver of TAPVR severity because it backs up blood in the pulmonary venous side while reducing effective systemic perfusion.



# Atrial Shunt Restriction

- RASD sweep: 2 → 12 mmHg/(L/min).
- Atrial shunt flow falls: 0.668 → 0.311 L/min.
- Systemic pressure falls: 66.55 → 31.26 mmHg.
- Aortic output falls: 0.675 → 0.303 L/min.
- $O_2$ delivery falls: 84.7% → 40.9% of normal.
- A restrictive ASD limits blood crossing RA → LA, reducing LV filling and systemic output.
- Overall conclusion: the model suggests increasingly anomalous pulmonary venous return worsens oxygenation and systemic output; obstruction causes the sharpest drop in pressure and flow.



# Final Conclusions

- PAPVR decreases systemic $O_2$ delivery to 93.7% of normal, showing a partial/anatomically milder version of anomalous return.
- Unobstructed TAPVR drops $O_2$ delivery to 67.9% of normal due to abnormal routing and atrial-level mixing.
- Obstructed TAPVR produces the largest decrease in systemic pressure, aortic output, and $O_2$ delivery while increasing Ppv.
- As ASD resistance increases, shunt flow and systemic output decrease, confirming the importance of atrial-level communication.
- Overall: the model links abnormal TAPVR anatomy to measurable physiology — pressure buildup, reduced flow, and reduced oxygen delivery.

# Validation, Limitations, and Future Work

Validation approach
• Using the baseline infant physiology which consists of HR 140 bpm and scaled flows/volumes
• Following physiological trend in which TAPVR reduces systemic delivery and the obstruction increases Ppv
• Literature consistency: atrial communication is required; obstruction worsens pulmonary venous pressure

Current limitations
• Simplified 0D model, not patient-specific geometry
• Right ventricle/pulmonary artery dynamics simplified
• Parameters are tuned for qualitative trends, not clinical prediction
• Does not model specific supracardiac/cardiac/infracardiac anatomy separately

Future improvements
• Calibrate against neonatal clinical ranges
• Add RV and pulmonary arterial compartments
• Map TAPVR subtypes to pathway resistance/anatomy
• Compare against patient-specific CFD or echocardiography/clinical literature

# References

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